MISB MOTION IMAGERY STANDARDS BOARD	MISB ST 0602.5
STANDARD **Annotation Metadata Set**	**24 February 2022**

1 Scope

This Standard documents KLV metadata to create and encode annotation data within a Motion Imagery data stream to allow for the dissemination and display of visual cues which enhance the exploitation of MISP-conformant Motion Imagery data.

2 References

[1] SMPTE ST 336:2017 Data Encoding Protocol Using Key-Length-Value.

[2] MISB ST 0107.5 KLV Metadata in Motion Imagery, Oct 2021.

[3] ISO/IEC 13818-1:2019 Information technology - Generic coding of moving pictures and associated audio information: Systems.

[4] MISB ST 1402.2 MPEG-2 Transport Stream for Class 1/Class 2 Motion Imagery, Audio and Metadata, Oct 2016.

[5] MIL-STD-2301A Computer Graphics Metafile (CGM) Implementation Standard for the National Imagery Transmission Format Standard, 5 Jun 1998.

[6] ISO/IEC 10918-1:1994 Information technology — Digital compression and coding of continuous-tone still images: Requirements and guidelines.

[7] ISO/IEC 15948:2004 Information technology — Computer graphics and image processing — Portable Network Graphics (PNG): Functional specification.

[8] W3C, "Scalable Vector Graphics (SVG) 1.1 (Second Edition)," [Online]. Available: https://www.w3.org/TR/SVG11/.

[9] MISB ST 0807.25 MISB KLV Metadata Registry, Jun 2020.

[10] SMPTE RP 210v13:2012 Metadata Element Dictionary.

3 Revision History

Revision	Date	Summary of Changes
0602.5	02/24/2022	• Added MIME data type image/svg+xml • Changed “element” to “item” per SMPTE usage • Editorial changes for readability; added acronyms • Updated references

4 Definitions and Acronyms

AOI	Area of Interest
BMP	Bit Map Image File
CGM	Computer Graphics Metafile
DIB	Device Independent Bitmap
IETF	Internet Engineering Task Force
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
JPG	Joint Photographic Expert Group
KLV	Key-Length-Value
MISB	Motion Imagery Standards Board
MIME	Multipurpose Internet Mail Extension
PNG	Portable Network Graphics
RP	Recommended Practice
SMPTE	Society of Motion Picture and Television Engineers
ST	Standard
SVG	Scalable Vector Graphics
VDC	Virtual Device Coordinate
W3C	World Wide Web Consortium

5 Introduction

Motion Imagery annotation is the process of visibly providing added information about objects appearing in a Motion Imagery data stream. An annotation consists of data which describes an object and is associated with the subset of frames for which the annotation is valid. A typical use case is adding annotation to objects within a MPEG-2 Transport Stream carrying the Motion Imagery. Such an application requires adding a new or using an existing elementary data stream – which this document calls a Motion Imagery annotation data stream – within a MPEG-2 Transport Stream.

SMPTE ST 336 [1] defines a Key-Length-Value (KLV) metadata encoding protocol for representing data items and data groups independent of the application or data transport method used. MISB ST 0107 [2] provides a set of baseline requirements for applying KLV metadata in Motion Imagery. ISO/IEC 13818-1 [3] in concert with MISB ST 1402 [4] describe the rules and requirements for managing MPEG-2 Transport Stream elementary streams.

Requirement(s)	
ST 0602.4-01	Motion Imagery annotation data shall be implemented as SMPTE ST 336 conformant Universal Sets.
ST 0602.4-02	Annotation private data streams shall be multiplexed in accordance ISO/IEC 13818-1 and MISB ST 1402.
ST 0602.4-03	All metadata shall be expressed in accordance with MISB ST 0107.

6 Motion Imagery Annotation Metadata

6.1 Preface Metadata Items

A Motion Imagery annotation data stream is comprised of a sequence of KLV metadata items. The KLV items *Byte Order*, *Active Lines per Frame* and *Active Samples per Line* are ancillary preface metadata items which provide information to aid receivers in interpreting the Annotation Universal Set data that follows. Both Active Lines per Frame and Active Samples per Line specify the number of lines (or rows) and the number of samples (or columns) respectively of the Motion Imagery encoded in the compression process. Frequent inclusion of these items enhances random accessibility. Note that the Active Lines per Frame (i.e., the rows) and Active Samples per Line (i.e., the columns) are the pixel array dimensions of the native Motion Imagery data prior to compression.

Requirement(s)	
ST 0602.4-04	When the metadata item Byte Order has not been inserted within an annotation data stream in the previous quarter second of program timeline, it shall be inserted into the annotation data stream immediately preceding an Annotation Universal Set.
ST 0602.4-05	When the metadata item Active Lines per Frame has not been inserted within an annotation data stream in the previous quarter second of program timeline, it shall be inserted into the annotation data stream immediately preceding an Annotation Universal Set.
ST 0602.4-06	When the metadata item Active Samples per Frame has not been inserted within an annotation data stream in the previous quarter second of program timeline, it shall be inserted into the annotation data stream immediately preceding an Annotation Universal Set.

6.2 Annotation Universal Metadata Set

The Annotation Universal Metadata Set 16-byte key follows the preface metadata items.

Requirement	
ST 0602.4-07	The Annotation Universal Metadata Set key shall be present.

Both the Locally Unique Identifier and the Event Indication items need to be present to signal the annotated object and the state of the annotation respectively.

6.2.1 Locally Unique Identifier

The Locally Unique Identifier item identifies the object with which the annotation message is associated.

Requirement	
ST 0602.4-08	The Annotation Metadata Set shall include the Locally Unique Identifier item.

6.2.2 MIME Media Type

The MIME Media Type item describes the MIME encoding type of the accompanying MIME Data.

6.2.3 MIME Data

The MIME Data item contains the MIME encoded data of the annotation message. Table 1 lists the allowed MIME media types and MIME Data Reference.

Table 1: Allowed MIME Media Types and Data Reference

MIME Media Type	MIME Data Reference
image/x-ms-bmp	Device Independent Bitmap (DIB or BMP) file format
image/cgm[1]	MIL-STD-2301A [5]
image/jpeg	ISO 10918-1 [6]
image/png	ISO/IEC 15948 [7]
image/svg+xml	https://www.w3.org/TR/SVG 1.1/ [8]. MISB recommends absolute (i.e., not relative) sizing only and SVG static-only features (i.e., no external content, scripting, animations, etc.)

Requirement(s)	
ST 0602.4-10	Encoders shall use only MIME media types as listed in MISB ST 0602.
ST 0602.4-11	A decoder shall be capable of decoding all MIME Media Types listed in MISB ST 0602.

6.2.4 X and Y Viewport Position in Pixels

The X and Y Viewport Position in Pixels item indicates the location of the MIME Data reference point, typically the image origin, defined independently for each data type, per Table 2. The reference for the X and Y position is the (0,0) origin in the upper-left corner of the original essence data image.

[1] Versions of this specification prior to version 2 referred to the MIME type for CGM as "cgm", rather than the MIME-standard format of "image/cgm". Decoder implementations encountering streams produced according to these prior versions should treat a MIME type of "cgm" the same as those with MIME type of "image/cgm". Encoders producing streams in accordance with this specification are not to use "cgm" for the MIME media type of any CGM annotation.

Table 2: X and Y Viewport Position in Pixels

MIME Type	Interpretation of X and Y Viewport Position in Pixels
image/x-ms-bmp	Position of lower left corner of BMP within viewport
image/cgm	Origin of the CGM's VDC (Virtual Device Coordinate) within the viewport
image/jpeg	Position of upper left corner of JPEG within viewport
image/png	Position of upper left corner of PNG within viewport
image/svg+xml	Position of upper left corner of SVG within viewport

6.2.5 Event Indication

The Event Indication item characterizes the event as initiated (i.e.., NEW) or modifying (i.e., MOVE, MODIFY, STATUS, DELETE) the message. An annotation for an object has persistence for a period of 20 seconds, where after that time an implementation may consider the annotated object deleted. Events for an annotation object apply to a particular frame or frames of the Motion Imagery. Table 3 lists the states for Event Indication.

Table 3: Event Indications

Event Indication	Action Effected
NEW	Creates/initiates an annotated object
MOVE	Updates an annotated object's position within the viewport
MODIFY	Changes to the annotated object's data
STATUS	Periodic message that describes the current contents and location of the annotated object
DELETE	Indicates the termination of the associated annotated object

Requirement(s)	
ST 0602.4-09	The Annotation Metadata Set shall include the Event Indication item with a specific event message as defined in MISB ST 0602 Table 3.
ST 0602.4-12	A "NEW", "MODIFY", or "STATUS" event shall include the MIME Media Type, the MIME Data, the Modification History, the X Location, the Y Location, and the Z-Order items.
ST 0602.4-13	A "MOVE" event shall include the X Location, Y Location, and Z-Order items.
ST 0602.4-14	A "DELETE" event shall include the Modification History item.
ST 0602.4-15	An Annotation Source key shall be present during a "NEW" event.
ST 0602.4-16	An Annotation Source key shall be present during a "STATUS" event.
ST 0602.4-17	To support random access, at least one "STATUS" or "MODIFY" event shall be provided for each annotated object, at least once every five seconds.

6.2.6 Modification History

The Modification History item provides information identifying the author of the most recent event such as "NEW", "MODIFY" and "DELETE. The specific contents are user definable.

6.2.7 Annotation Source

The Annotation Source item describes the source of the annotation and is only required for "NEW" and "STATUS" events. Table 4 lists the allowed Annotation Source values.

Table 4: Definition of Annotation Source Values

Value	Definition
0x00000000	Manually Annotated
0x00000001	Automated from the Basic Encyclopedia or other sources
0x00000002	Automated from user defined latitude / longitude center point
0x00000004	Automated from user defined AOI (Area of Interest)
0x00000008	Automated from pixel analysis

6.2.8 Z-Order

The Z-Order item defines the rendering order of the annotations, which are from lowest to highest Z-Order in a given frame.

Requirement	
ST 0602.4-18	Annotations without a Z-Order value, such as those created based on previous versions of this document, shall be treated as having a Z-Order of zero (Z-Order = 0) for rendering purposes.

7 Annotation Metadata Set

The MISB ST 0807 [9] registry defines the Annotation Metadata Set UL (i.e., key) and all metadata items.

Metadata Key	Name	Definition	Type	Value Length (bytes)	Value Range	Usage
06.0E.2B.34.01.01.01.01.03.01.02.01.02.00.00.00 (CRC 21119)	Byte Order	Byte order of the metadata	int16	2	Fixed value = 0x4D4D	Ref Section 6.1
06.0E.2B.34.01.01.01.01.04.01.03.02.02.00.00.00 (CRC 15893)	Active Lines per Frame	Total number of active lines rows) in a frame of an image matrix	uint16	2		Ref Section 6.1
06.0E.2B.34.01.01.01.01.04.01.05.01.02.00.00.00 (CRC 23846)	Active Samples per Line	Total number of active samples (columns) in a line of an image matrix	uint16	2		Ref Section 6.1
06.0E.2B.34.02.01.01.01.0E.01.03.03.01.00.00.00 (CRC 39803)	Annotation	Annotation Universal Metadata Set		V		All Object Messages
06.0E.2B.34.01.01.01.01.01.03.03.01.00.00.00.00 (CRC 5618)	Locally Unique Identifier	A 4-byte locally unique ID	uint32	4		All Object Messages

Metadata Key	Name	Definition	Type	Value Length (bytes)	Value Range	Usage
06.0E.2B.34.01.01.01.01.03.02.01.06.03.00.00.00 (CRC 21885)	(Media) Description	Freeform textual description (per SMPTE RP 210 [10]) providing title or text description of enclosed MIME data	utf8	V	Free text	NEW MODIFY STATUS
06.0E.2B.34.01.01.01.07.04.09.02.00.00.00.00.00 (CRC 7035)	MIME Media Type	MIME media type	utf8	V	Ref Table 1	NEW MODIFY STATUS
06.0E.2B.34.01.01.01.01.0E.01.02.05.01.00.00.00 (CRC 5825)	MIME Data	MIME encoded data of annotation message	Opaque	V	Ref Table 1	NEW MODIFY STATUS
06.0E.2B.34.01.01.01.01.07.01.02.03.01.00.00.00 (CRC 33357)	X Viewport Position in Pixels	X position of an object within a viewed image	int16	2		NEW MOVE MODIFY STATUS
06.0E.2B.34.01.01.01.01.07.01.02.03.02.00.00.00 (CRC 6545)	Y Viewport Position in Pixels	Y position of an object within a viewed image	int16	2		NEW MOVE MODIFY STATUS
06.0E.2B.34.01.01.01.01.05.01.01.02.00.00.00.00 (CRC 8174)	Event Indication	Describes the event	utf8	1 (32 bytes max)	0x31 = NEW[2] 0x32 =MOVE 0x33 =MODIFY 0x34 =DELETE 0x35 =STATUS	All Object Messages
06.0E.2B.34.01.01.01.01.0E.01.02.05.02.00.00.00 (CRC 36125)	Modification History	Identification of most recent significant event's author	utf8	V	Free text	NEW MODIFY DELETE STATUS
06.0E.2B.34.01.01.01.01.0E.01.02.05.03.00.00.00 (CRC 64425)	Annotation Source	Source of the specified annotation object	int32	4		NEW STATUS
06.0E.2B.34.01.01.01.01.0E.01.02.05.06.00.00.00 (CRC 18412)	Z-Order	Number defining the drawing order of annotations in a frame	BER-OID	V		NEW MOVE MODIFY

[2] For example, a NEW event would have a value of 0x31 (i.e., the one-byte string "1"), a MOVE event would have the value 0x32 ("2"), etc.